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MSA | SPC | Control Plan

Quality Intelligence Platform

Integrated MSA · SPC · Control Plan Solution
for IATF 16949 & OEM Compliance

Measurement · Control · Intelligence

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MSA Module

Gauge & Instrument Master

- Centralized gauge registry
- Variable / attribute classification
- Calibration status & due-dates
- Measurement range & accuracy
- Instrument usage history

MSA Study Management

- GR&R (AIAG MSA 4th / 5th edition)
- Repeatability & Reproducibility
- %GR&R & ndc calculation
- Attribute MSA: Bias, Linearity
- Stability analysis

Study Configuration

- Appraiser selection & roles
- Part quantity & trial definition
- Measurement method selection
- OEM-specific study plan templates
- Configurable study workflows

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Automated Analysis & Traceability

Automated Analysis & Results

- Automatic GR&R calculation
- Graphical outputs: Xbar-R, Range, Contribution %
- Accept / Conditional / Reject decision logic
- Trend analysis across studies
- Historical comparison of MSA results

MSA Compliance & Traceability

- Link MSA → Control Plan → Process → Machine → Part
- Study approval workflow
- Customer & auditor-ready reports
- Re-study triggers on gauge change / ECN
- Full revision history & traceability

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SPC Module

Real-Time SPC Monitoring

- Live data capture (manual, machine, IoT, MES)
- Auto-sampling based on control plan
- Multi-parameter process monitoring
- Real-time quality alerts

SPC Chart Types

- X-bar & R / X-bar & S
- Individuals & Moving Range (I-MR)
- Attribute: P, NP, C, U charts
- Custom OEM-specific chart rules

Process Capability Analysis

- Cp, Cpk, Pp, Ppk calculation
- Short-term vs long-term capability
- Capability trend monitoring
- Customer-specific capability criteria

Control Rules & Alarms

- Western Electric & AIAG SPC rules
- Configurable alarm thresholds
- Escalation workflows (email, dashboard)
- MES alert integration

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Analytics & Integration

SPC Analytics

- Out-of-control analysis
- Pareto of causes
- Shift-wise & machine-wise performance
- Process drift detection
- Cost of Poor Quality (COPQ) linkage

SPC Integration

- Machine data: PLC, SCADA
- MES / aMES systems
- ERP quality & production modules
- Supplier SPC for incoming quality
- IoT-enabled real-time streams

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Control Plan Module

Creation & Management

- Prototype / Pre-launch / Production plans
- AIAG & OEM standard templates
- Operation-wise control definition
- Process flow integration

Characteristics Definition

- Product characteristics (Critical/Major/Minor)
- Process parameters: CTQ, CTC
- Measurement method & gauge linkage
- Sample size & frequency

Reaction Plan Management

- Out-of-control reaction steps
- Responsibility assignment
- Escalation triggers
- CAPA linkage

Version & Change Mgmt

- Revision control with approval workflow
- Change impact analysis
- Historical control plan comparison
- Customer re-approval support

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Execution & Integration

Shop Floor Execution

- Digital control plan execution
- Operator-guided inspection steps
- Real-time compliance checks
- Deviation alerts

Integrated Quality Linkage

- Direct linkage: PFMEA ↔ MSA ↔ SPC ↔ Inspection
- Auto-update control plan on ECN / FMEA change
- Revision control with approval workflow
- Change impact analysis
- Customer re-approval support

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Cross-Module Intelligence

One-Click Traceability Chain



Automatic MSA Revalidation

When SPC drifts detected, system auto-triggers MSA revalidation to ensure measurement system integrity.

SPC Auto-Blocking

On control plan violation, SPC automatically blocks the process preventing non-conforming output.

Quality Health Scoring

Dashboard-based quality health scoring across all modules providing a single-pane-of-glass view.

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Dashboards & KPIs

GR&R Acceptance Rate

Process Capability Compliance

Out-of-Control Frequency

Control Plan Adherence

Customer PPM Impact

All KPIs available on real-time dashboards with drill-down capability and customer-ready export.

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Business Benefits

Reduced Variation & Scrap

Real-time SPC monitoring catches drift early, reducing rework and scrap costs significantly.

Faster OEM Approvals

AIAG-compliant, auditor-ready reports accelerate customer sign-off processes.

Zero Manual SPC Spreadsheets

Fully automated data capture from machines, MES, and IoT eliminates manual entry.

Audit-Ready MSA & Control Plans

Complete traceability and approval workflows satisfy IATF 16949 audits effortlessly.

IATF 16949 Compliance

Built-in compliance framework ensures continuous adherence to quality management standards.

Improved Shop-Floor Discipline

Operator-guided digital control plan execution ensures consistent process adherence.

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One Platform.
Total Quality Intelligence.

MSA · SPC · Control Plan · IATF 16949

Connecting measurement, control, and analytics
into a single, integrated quality ecosystem.